Mathematical Foundation of ML

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Problem Statement

Show that for any $0 < \varepsilon < 1$, there exists a neural network $\tilde{D}$ with $\mathcal{O}\left(\log^2\left(\varepsilon^{-1}\right)\right)$ weights that approximates the function $D(x_1, x_2) = e^{x_1} x_2$, such that

$$\left\| D - \tilde{D} \right\|_{L_\infty([-1,1]^2)} \leq \varepsilon.$$

You may use the following results: - For any $n \geq 1$, there exists a neural network $D_{1,n}$ with $\mathcal{O}(n)$ weights such that

$$\max_{-1 \leq x \leq 1} |e^x - D_{1,n}(x)| \leq c_1 e^{-n}.$$

- For any $n \geq 1$, there exists a neural network $D_{2,n}$ with $\mathcal{O}(n)$ weights such that

$$\max_{-2e \leq x_1, x_2 \leq 2e} |x_1 x_2 - D_{2,n}(x_1, x_2)| \leq c_2 e^{-\sqrt{n}}.$$

Solution

Step 1: Approximation of e^{x_1}

The function we want to approximate is $D(x_1, x_2) = e^{x_1} x_2$, which involves the product of two terms: e^{x_1} and x_2.

We begin by approximating the exponential function e^{x_1}. According to the hint, for any $n \geq 1$, there exists a neural network $D_{1,n}(x_1)$ with $\mathcal{O}(n)$ weights that satisfies:

$$\max_{-1 \leq x \leq 1} |e^{x_1} - D_{1,n}(x_1)| \leq c_1 e^{-n}.$$

This result tells us that we can approximate e^{x_1} arbitrarily well on the interval $[-1, 1]$ using a neural network $D_{1,n}$, with the approximation error decreasing exponentially in n.

To ensure that the error of the approximation is less than ε, we choose $n = \mathcal{O}(\log(\varepsilon^{-1}))$. This choice guarantees that:

$$\max_{-1 \leq x_1 \leq 1} |e^{x_1} - D_{1,n}(x_1)| \leq \varepsilon.$$

Step 2: Approximation of x_1x_2

Next, we need to approximate the product $e^{x_1}x_2$. Since $D_{1,n}(x_1)$ is our approximation for e^{x_1}, we now focus on approximating the product of $D_{1,n}(x_1)$ and x_2.

From the hint, we know that for any $n \geq 1$, there exists a neural network $D_{2,n}(x_1, x_2)$ with $\mathcal{O}(n)$ weights such that:

$$\max_{-2e \leq x_1, x_2 \leq 2e} |x_1x_2 - D_{2,n}(x_1, x_2)| \leq c_2 e^{-\sqrt{n}}.$$

We apply this result to approximate the product $e^{x_1}x_2$, noting that e^{x_1} lies in the range $[\frac{1}{e}, e]$ for $x_1 \in [-1, 1]$ and $x_2 \in [-1, 1]$. Therefore, the product $e^{x_1}x_2$ lies within the interval $[-2e, 2e]$, which fits within the domain of the approximation provided by $D_{2,n}(x_1, x_2)$.

To ensure that the error of the product approximation is less than ε, we choose $n = \mathcal{O}(\log^2(\varepsilon^{-1}))$, ensuring:

$$\max_{-1 \leq x_1, x_2 \leq 1} |D(x_1, x_2) - D_{2,n}(D_{1,n}(x_1), x_2)| \leq \varepsilon.$$

Step 3: Constructing the Neural Network $\tilde{D}$

We can now construct the neural network $\tilde{D}$ as a composition of the two neural networks $D_{1,n}$ and $D_{2,n}$, where $D_{1,n}$ approximates e^{x_1} and $D_{2,n}(D_{1,n}(x_1), x_2)$ approximates the product $e^{x_1}x_2$.

The overall error from this construction is bounded by ε, ensuring:

$$\left\| D(x_1, x_2) - \tilde{D}(x_1, x_2) \right\|_{L_\infty([-1,1]^2)} \leq \varepsilon.$$

Step 4: Bounding the Number of Weights

From the given hint: - The neural network $D_{1,n}$ has $\mathcal{O}(n)$ weights, where $n = \mathcal{O}(\log(\varepsilon^{-1}))$. - The neural network $D_{2,n}$ has $\mathcal{O}(n)$ weights, where $n = \mathcal{O}(\log^2(\varepsilon^{-1}))$.

Thus, the total number of weights required in $\tilde{D}$ is $\mathcal{O}(\log^2(\varepsilon^{-1}))$.

Conclusion

Thus, for any $0 < \varepsilon < 1$, there exists a neural network $\tilde{D}$ with $\mathcal{O}(\log^2(\varepsilon^{-1}))$ weights such that

$$\left\| D(x_1, x_2) - \tilde{D}(x_1, x_2) \right\|_{L_2([-1,1]^2)} \leq \varepsilon.$$